

COIMBATORE INTL, India

WMO: 433210

Lat: 11.031N

Lon: 77.044E

Elev: 1324

StdP: 14.01

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	65.2	66.9	41.7	40.9	90.1	46.4	48.8	87.5	14.0	81.9	12.6	81.5	1.2	20	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.5	98.1	72.0	96.5	72.0	94.8	72.6	78.2	88.3	77.4	87.1	77.0	86.5	7.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.0	142.9	81.0	75.4	139.8	80.6	74.8	137.1	80.2	42.8	89.0	42.1	86.9	41.5	86.7	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.5	14.9	DB	61.7	100.4	3.0	1.7	59.5	101.6	57.8	102.7	56.2	103.6	54.0	104.9	(4)
(5)			WB	56.0	80.9	2.0	1.8	54.6	82.3	53.4	83.3	52.3	84.4	50.9	85.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.8	77.1	80.3	84.2	86.0	84.4	81.4	80.1	80.3	81.2	80.3	78.2	76.4		
	DBStd	3.58	2.01	2.41	2.36	2.35	2.66	2.59	2.05	1.96	1.74	2.22	2.22	2.20		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11249	841	847	1059	1081	1065	943	933	940	935	941	845	819		
	CDD65	5774	376	427	594	631	600	493	468	475	485	476	395	354		
	CDH74	53798	3003	4350	7047	7647	6695	4427	3674	3680	4054	3888	2896	2437		
	CDH80	22373	989	1981	3675	3946	3119	1683	1246	1298	1556	1412	834	634		
(14) Wind	WSAvg	6.8	4.4	5.1	5.4	5.9	7.9	10.7	10.7	10.0	8.7	5.3	3.7	4.0		
(15) Precipitation	PrecAvg	25.5	0.2	0.3	0.5	1.5	3.1	1.2	1.7	1.0	2.3	6.3	5.0	2.2		
	PrecMax	52.7	2.1	4.1	4.3	3.7	6.7	4.1	5.7	5.0	7.4	15.6	27.0	8.9		
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.8	0.0	0.2	0.0	0.0	0.5	1.2	0.0		
	PrecStd	9.0	0.5	0.8	1.0	0.9	1.8	1.1	1.4	1.1	2.0	3.9	5.2	2.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.8	96.0	99.0	100.5	98.4	95.0	91.7	91.8	93.2	92.7	89.9	88.6		
		MCWB	67.9	67.5	68.8	72.5	75.5	75.9	74.6	75.2	74.3	73.1	71.6	68.9		
	2%	DB	88.1	93.2	97.2	98.6	96.2	92.5	89.7	90.0	91.3	91.0	87.9	86.4		
		MCWB	67.3	67.9	68.9	72.9	76.0	75.4	74.5	74.7	74.2	73.2	71.9	69.0		
	5%	DB	86.8	91.5	95.8	96.9	94.4	90.3	88.0	88.4	89.7	89.3	86.3	85.2		
		MCWB	67.2	67.2	69.3	72.9	76.1	75.0	74.1	74.4	74.1	73.4	71.8	68.8		
	10%	DB	85.4	89.7	94.1	95.0	92.6	88.2	86.4	86.8	87.9	87.4	84.6	83.7		
		MCWB	67.1	67.5	69.7	73.5	76.1	74.6	73.9	74.3	74.0	73.5	71.6	68.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.3	74.7	77.3	78.6	79.6	78.3	76.6	77.3	77.8	77.1	76.5	74.8		
		MCDB	81.7	82.2	86.7	87.5	90.6	89.0	87.0	87.2	86.3	84.7	82.8	80.9		
	2%	WB	72.2	73.6	76.2	77.7	78.5	77.1	75.8	76.3	76.4	76.1	75.4	73.6		
		MCDB	80.2	82.3	85.1	86.7	89.4	87.5	85.7	85.9	85.3	83.8	81.8	79.6		
	5%	WB	71.3	72.7	75.4	77.2	77.8	76.3	75.2	75.5	75.6	75.5	74.6	72.8		
		MCDB	79.4	81.8	84.1	86.2	88.1	86.0	84.8	84.7	84.5	83.2	80.7	78.6		
	10%	WB	70.5	71.9	74.7	76.6	77.2	75.6	74.6	74.9	74.9	74.8	74.0	72.0		
		MCDB	79.0	81.1	83.5	85.6	87.1	84.5	83.1	83.4	83.5	82.3	79.8	77.6		
(35) Mean Daily Temperature Range	MDBR		17.7	20.0	20.9	19.5	17.4	14.5	14.0	14.5	15.6	14.6	13.4	15.3		
	5% DB	MCDBR	19.2	21.9	22.5	20.9	19.4	16.9	16.1	16.5	17.6	17.2	16.0	17.3		
		MCWBR	5.3	6.2	7.0	5.9	4.4	4.2	4.1	3.9	3.9	4.4	4.5	4.9		
	5% WB	MCDBR	15.8	18.5	20.7	19.5	17.9	15.4	14.9	14.9	15.5	14.6	12.7	13.1		
		MCWBR	5.0	5.5	6.7	5.7	5.2	4.7	4.4	4.3	4.4	4.5	4.3	4.4		
(40) Clear-Sky Solar Irradiance	taub		0.446	0.456	0.517	0.552	0.541	0.470	0.491	0.475	0.454	0.508	0.488	0.451		
	taud		2.184	2.163	2.011	1.964	2.011	2.256	2.177	2.232	2.295	2.108	2.139	2.209		
	Ebn at Noon		272	274	259	247	245	260	256	264	272	256	258	267		
	Edh at Noon		46	49	58	60	56	43	47	45	43	51	48	44		
(44) All-Sky Solar Radiation	RadAvg		1708	1938	2012	1937	1817	1524	1446	1515	1652	1513	1414	1486		
	RadStd		73	82	91	72	112	106	110	98	123	83	118	98		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+1.17	+1.13	N/A	+2.00	+0.59	+0.54	N/A	N/A	N/A
(47) Regional (3 neighbors)	DBAvg	+0.81	N/A	N/A	+1.84	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page